MIL-W-14718A (MU)
2 January 1969
Superseding
MIL-W-14718
17 April 1968

MILITARY SPECIFICATION

WARHEAD, GUIDED MISSILE, TRAINING, XM237E1, LOADING, ASSEMBLING AND PACKING

1. SCOPE

1.1 This specification covers the quality assurance provisions for loading, assembling and packing of warhead designated as Warhead, Guided Missile, Training, XM237E1.

2. APPLICABLE DOCUMENT

2.1 The following documents of the issue in effect on date of invitation for bids form a part of this specification to the extent specified herein.

SPECIFICATIONS

MILITARY

MIL-D-1000 - Drawing, Engineering and Associated List.

STANDARDS

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-1167 - Ammunition Data Cards.
MIL-STD-1168 - Lot Numbering of Ammunition.
MIL-STD-1235 - Single and Multilevel Continuous Sampling Procedures and Tables for Inspection by Attributes.

FSC: 1336

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DRAWINGS

US ARMY MUNITIONS COMMAND

9211595 - Crate, Packing, Ammo, Dual for Warhead, GM, HE, XM155 and Training, XM237E1.
9211596 - Crate, Packing, Ammo, Single for Warhead, GM, HE, XM155 and Training, XM237E1.
9233456 - Crate, Packing, Ammo, Dual, Modified for Warhead, GM, HE, XM155, and Training, XM237E1.
9226301 - Warhead, Guided Missile, Training, XM237E1, Loading Assembly and Marking.

PUBLICATIONS

US ARMY MUNITIONS COMMAND

ET-9226301 - Equipment Tabulation.

(Copies of specifications, standards, drawings and publications required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer).

3. REQUIREMENTS

3.1 Material. - Materials shall be in accordance with applicable drawings and specifications.

3.2 Warhead assemblies. - The warhead assemblies shall comply with all requirements specified on Drawing (dwg.) 9226301 and with all requirements specified in applicable specifications.

3.3 Painting and marking. - Painted surfaces and marking shall be restored by the loading contractor when such surfaces or marking have been damaged or obliterated.

3.4 Workmanship. - Loading and assembly shall be accomplished in a thorough, workmanlike manner. All parts shall be free of any foreign matter. The cleaning method used shall not be injurious to any part nor shall the parts be contaminated by the cleaning agent. Surface coatings shall be continuous except for a few light scratches not exposing base material. All required marking shall be neat and legible.

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4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified therein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Submission of product. - At the time each completed lot of items deliverable under the contract is submitted to the Government for acceptance, the contractor shall supply the following information accompanied by a certificate which attests that the information provided is correct and applicable to the product being submitted:

a. A statement that the lot complies with all of the quality assurance provisions specified in this specification.

b. Drawing and specification number and date, together with identification and date of changes thereto.

c. A statement that all material purchased by the contractor meets requirements, when such material is controlled by Government or commercial specifications referenced in any of the contractual documents, and that certificates of conformance are on file and available for review.

d. Number of items in the lot.

e. Date submitted.

The certificate shall be signed by a responsible agent of the certifying organization. The initial certificate submitted shall be substantiated by evidence of the agent's authority to bind his principal. Substantiation of the agent's authority will not be required with subsequent certificates unless, during the course of the contract, this authority is vested in another agent of the certifying organization.

4.2 Inspection Provisions.

4.2.1 Lot formation. - The term "inspection lot" as used in this specification is defined as an essentially homogeneous collection of units of product from which a representative sample is drawn and inspected to determine conformance with applicable requirements. The sample selected shall represent only that quantity of units from which the sample was drawn and shall not be construed to represent any prior or subsequent quantities presented for inspection. Homogeneity shall be considered to exist provided the inspection lot has been produced by one manufacturer, in one

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unchanged process, in accordance with the same drawing, same drawing revision, same specification and same specification revision and complies with the provisions for submission of product as specified in MIL-STD-105. Changes to the process, specification, or drawing not affecting safety, performance, interchangeability, or storage, as determined by the Government, shall not be deemed to alter the homogeneity of an inspection lot. All material submitted for inspection in accordance with this specification shall comply with the homogeneity criteria specified herein regardless of the type of sampling procedure which is being applied to determine conformance with requirements. Lot numbering, as required, shall be in accordance with MIL-STD-1168.

4.2.2 Examination. - The examinations listed in the following subparagraphs shall be performed on inspection lots (see 4.2.1) of the components or assemblies designated in the applicable paragraph headings. Inspection for critical defects (and major defects, when so specified) shall be 100 percent. Sampling plans for major and minor defects shall be in accordance with MIL-STD-105 except that continuous sampling plans in accordance with MIL-STD-1235 may be used if approved by the Government. Instead of applying the AQL's designated herein to the groups of major and minor defects listed, the contractor may elect to apply AQL 's to individual defects using an AQL of 0.40 percent for each major defect and an AQL of 0.65 percent for each minor defect. When this method is elected, it shall be applied to all of the major and minor defects listed in the applicable subparagraph of this section, except where 100 percent inspection is specified, and be documented as part of the contractor's inspection system. Equipment necessary for the performance of the inspections listed shall be in accordance with 4.3.4.

4.2.2.1 Warhead, prior to loading (see dwg. 9230608).

Categories	Defects	Method of Inspection	Code No. (see 6.2)
Critical:	None defined.		
Major:	AQL 0.40 percent		
101.	Body cracked or otherwise damaged	Visual	01001
102.	Insert damaged	Visual	01002
Minor:	AQL 0.65 percent		
201.	Evidence of poor workmanship (see 3.4)	Visual	01004

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4.2.2.2 Assembly, prior to bonding cover (see dwg. 9226301).

Categories	Defects	Method of Inspection	Code No.
Critical: None defined.			
Major: None defined.			
Minor:	AQL 1.00 percent		
201.	Cavity not filled	Visual	02001
202.	Funnel mark not leveled	Visual	02002

4.2.2.3 Loading assembly (see dwg. 9226301).

Categories	Defects	Method of Inspection	Code No.
Critical: None defined.			
Major:	AQL 1.00 percent		
101.	Total length, max	Gage	03001
102.	Outside diameter, max	Gage	03002
103.	Cover insecurely bonded	Visual/ Manual	03003
104.	Paint in threads, safety and arming cavity or aft cup	Visual	03004
105.	Aft cup loose or damaged	Visual	03005
106.	Simulated booster missing, loose or above flush	Visual/ Manual	03006
Minor:	AQL 1.00 percent		
201.	Marking and data misleading or unidentifiable	Visual	03007
202.	Evidence of poor workmanship (see 3.4)	Visual	03008

4.2.2.4 Shipping container, prior to sealing, with warhead (see dwg. 9211595, dwg. 9211596, or dwg. 9233456 whichever applies).

Categories	Defects	Method of Inspection	Code No.
Critical: None defined.			

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Major:	AQL 0.65 percent		
101.	Holding plate missing or loose...	Visual	04001
102.	Warhead not seated properly.....	Visual	04002
103.	Hold down strap missing, broken or insecure.....	Visual/ Manual	04003
Minor:	AQL 1.00 percent		
201.	Foreign matter inside crate.....	Visual	04004
202.	Strapping improperly assembled...	Visual	04005

4.2.2.5 Shipping container, sealed (see dwg. 9211595, dwg. 9211596, or dwg. 9233456 whichever applies).

Categories	Defects	Method of Inspection	Code No.
Critical:	None defined.		
Major:	AQL 1.00 percent		
101.	Strapping missing, broken or loose.....	Visual/Manual	05001
102.	Board broken, split or curled...	Visual	05002
103.	Skid missing or damaged excessively.....	Visual	05003
104.	Contents exposed.....	Visual	05004
105.	FSC number misleading or unidentifiable.....	Visual	05005
Minor:	AQL 1.50 percent		
201.	Fastener missing or loose....	Visual/Manual	05006
202.	Strapping improperly assembled..	Visual	05007
203.	Nail protruding.....	Visual	05008
204.	Marking missing, misleading or unidentifiable.....	Visual	05009

4.2.3 Testing

4.2.3.1 Assembly weight (see dwg. 9226301) - Major defect - Code No. 03009. - This determination shall be made 100 percent using a scale accurate to within 0.01 pound. Any assembly failing to comply with the drawing requirement shall be classed defective and removed from the lot.

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4.3.4 Inspection Equipment - Equipment Tabulation number ET-9226301 identifies the inspection equipment required to perform the examinations and tests prescribed in this section. The contractor shall design inspection equipment in accordance with the instructions in paragraph 6.3.

4.3.4.1 Government rights to documention. - Inspection equipment drawings and lists provided and revised in accordance with the requirements of the ET may be used by DOD activities for design, procurement, manufacture, testing, evaluation, production and receiving inspection, overhaul, shipping, storage, identification of stock, ordering and storage of replacement parts, inspection of items at overhaul, general maintenance of equipment, construction, survey and whenever inspection equipment drawings are needed.

5. PREPARATION FOR DELIVERY

5.1 Packing and marking.

5.1.1 Level C. The loaded warhead assemblies shall be packed and marked in accordance with dwg. 9211595, Dwg. 9211596, or dwg. 9233456 whichever applies.

5.2 Data cards. - Data cards shall be prepared for each lot (see 6.1).

6. NOTES

6.1 Ordering data. - Procurement documents shall specify the following:

a. Title, number and date of this specification.

b. Data cards. - Data cards shall be prepared for each lot in accordance with the information specified in Standard MIL-STD-1167.

6.2 Inspection code numbers.- The five digit code numbers assigned to the inspections herein are to facilitate future data collection and analysis by the Government. These code numbers are used to correlate the characteristics cited on Equipment Lists with the inspections listed in this specification. They should also be cited as references on drawings of equipment designs submitted by the contractor (see 6.3.5).

6.3 Inspection equipment. - The contractor shall design inspection equipment as required by the referenced inspection Equipment Lists (EL) in accordance with the instructions of paragraphs 6.3.1 through 6.3.5.

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6.3.1 Inspection equipment lists. - Inspection equipment lists indicate the availability of inspection equipment designs by showing in the "number" column of the list of inspection equipment (Form SMUPA 1010) the numbers of drawings or Federal Stock numbers of existing equipment designs, or codes as indicated in paragraph 6.3.2. Design action required of the contractor is described in paragraphs 6.3.3 and 6.3.4. The contractor will be required to prepare detailed drawings in accordance with 6.3.4 for all the equipment coded as "Contractor Design" in the number column. These contractor designs must be approved by the Government prior to fabrication or procuring of the equipment. Designs shall be submitted for approval as specified in 6.3.5.

6.3.2 Inspection equipment list codes. - The inspection equipment as defined in 6.3.3 and 6.3.4 will be designated in the EL by the following codes:

CD - Contractor design on controlled contractor format and/or commercial equipment.
MU - Army design, mandatory for use.

6.3.3 Army designs. - Army designs are reflected on detailed drawings which completely depict all the information necessary for the fabrication of the item of inspection equipment. The contractor need provide no design when an Army design is listed for an item of inspection equipment. Army designs fall into two basic classifications: mandatory (designated "MU") and non-mandatory (indicated by drawing or Federal Stock Number). When an inspection equipment list references mandatory Army designs, the contractor shall comply with, and use these designs accordingly. The contractor may, however, in connection with non-mandatory designs, and with the approval of the Government, design alternate inspection equipment or use comparable commercial equipment to facilitate his operations. Such contractor prepared designs or commercial equipment selections must be approved by the Government prior to fabrication or procuring of the equipment. Designs shall be submitted for approval as specified in 6.3.5.

6.3.4 Contractor designs. - Contractor designs are designs of inspection equipment for which the Government has assigned design responsibility to the contractor. Contractor designs shall be supported by detailed drawings which depict all information necessary to completely fabricate, calibrate and operate an item of inspection equipment. This requires that the necessary views, dimensions, materials, finish, notes, operating and calibration instructions be properly depicted in accordance

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with approved practices to the extent that further calculation or clarification will not be required. Unless otherwise specified, contractor designs may be developed on the format the contractor normally employs in his equipment design procedure provided such format reflects the detail and information specified above, subject to the following controls: All submitted contractor designs shall conform to MIL-D-1000, Category E, Form 2. Legibility and reproducibility shall permit conventional making of clearly understandable, high contrast reproductions. Contractors shall submit one copy of final designs as a rolled set. Designs shall be submitted for approval as specified in 6.3.5.

6.3.5 Submission of contractor designs. - All submitted designs shall contain a reference to the applicable five digit Code Number contained in Section 4 of this specification. Unless otherwise specified on the EL, all designs of equipment for inspection of defects classified as critical and major shall be submitted for approval to the Commanding Officer, Picatinny Arsenal, ATTN: SMUPA-ND14. All other designs of inspection equipment shall be approved by the inspection element of the agency administering the contract; submission shall be as directed by the contracting officer. Partial submission of inspection equipment designs is permissible and encouraged. However, the Arsenal completion date for design review will be based on the date of the final submission of designs. Picatinny Arsenal design review will be accomplished normally within one month after receipt.

Notice. - When Government drawings, specifications, or other data are used for any purpose other than in connection with a definitely related Government procurement operation, the United States Government thereby incurs no responsibility nor any obligation whatsoever; and the fact that the Government may have formulated, furnished, or in any way supplied the said drawings, specifications, or other data is not to be regarded by implication or otherwise as in any manner licensing the holder or any other person or corporation, or conveying any rights or permission to manufacture, use, or sell any patented invention that may in any way be related thereto.

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Preparing activity:
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Project No. 1336-A-148

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